

LESSON 1 SQL AGGREGATE FUNCTIONS

1) Students Table

List all distinct departments in the students table

Syntax:

```
SELECT DISTINCT (DEPARTMENT)
FROM Students
```

Results:

Department

IT

HR

Finance

2) Get the average age of students per department

Syntax:

```
SELECT Student ID,  
AVG (age) as age of students
```

```
SELECT Department,  
AVG (AGE) AS avg age  
FROM Students  
Group by Department;
```

2) Results:

Department	Avg Age
IT	20,5
HR	22
Finance	23

3) Show departments with more than 1 student

```

SELECT Department,
       count(ID) as student_count
FROM Students
GROUP BY DEPARTMENT
HAVING count(ID) > 1 ;

```

Results

Department	student_count
IT	2
HR	2

4) Get all students whose age is between 21 and 23

```

SELECT Name
FROM Students
where age between 21 and 23 ;

```

Results:

5) List

SELECT

FROM
Where

Results:

S

Results:

Name	AGE
BOB	22
Charlie	21
Diana	23
Eve	22

5) List all students in the IT or HR department who are older than 21

```
SELECT Student_id, name,  
        age, department  
FROM Student  
Where 'IT' or HR age > 21 ;
```

Results :

Student_id	Name	AGE	DEPARTMENT
2	BOB	22	HR
	DIANA	23	
5	EVE	22	HR

6. Show ^{sum} total credits per department,
only for departments with more than
5 total credits

```
SELECT Department
      sum (credits) as tot_credits
FROM    COURSES
GROUP BY DEPARTMENT
HAVING tot_credits > 5;
```

RESULTS

DEPARTMENT	TOT_CREDITS
IT	

7. List all courses that do not have 4
credits

```
SELECT Course_name,
       credit
FROM    COURSES
WHERE   NOT CREDIT = 4;
```

RESULTS

COURSE NAME	CREDITS
IT	3
FINANCE	2
HR	3

8. Show the top 3 courses by credits in descending order

```
SELECT course_name, course_id,
       credits
FROM   courses
ORDER BY credits Desc
Limit 3;
```

RESULTS

COURSE_name	COURSE_ID	CREDITS
Python	102	4
Data Science	103	4
Statistics	105	3

9. Get the maximum, minimum, and average grade across all enrolments.

```
SELECT min (Grade) as min_Grade,
       max (Grade) as max_Grade,
       avg (Grade) as avg_Grade
FROM   ENROLLMENTS;
```

RESULTS

min_Grade	max_Grade	AVG_Grade
78	90	84.6

10) Count how many enrollments exist per course

```
SELECT COURSE_ID,
       count (*) as enrollment_count
FROM ENROLLMENTS
GROUP BY COURSE_ID;
```

RESULT

COURSE_ID	ENROLLMENT COUNT
101	1
102	1
103	1
104	1
105	1

11) Find total salary and total bonus per department

```
SELECT department,
       sum (salary) as totu Sal,
       sum (bonus) as totu bonus
FROM SALARIES
GROUP BY Department;
```

RESULTS

DEPARTMENT
IT
HR
FIN

12) Show is abo

SELECT

FROM
Where

RESULTS

RESULTS

DEPARTMENT	TOT. Salary	tot. Bonus
IT	122 000	10 500
HR	109 000	7500
FINANCE	70 000	6000

12) Show departments where average salary is above 55 000

```

SELECT DEPARTMENT,
        AVG (Salary) as avg. salary
FROM Salaries
WHERE avg (Salary) > 55 000
    
```

RESULTS

DEPARTMENT	AVG. SALARY
IT	60 000
Finance	70 000
IT	62 000

13) List employees whose salary plus bonus is greater than 60000

```
SELECT Employee_ID, name,
        Salary, bonus,
        sum (salary + bonus) as total_
        compensation
FROM Salaries
WHERE sum (salary + bonus) > 60000
```

RESULTS

Employee_ID	NAME	SALARY	BONUS	TOTAL Compensation
1	TOM	60000	5000	65000
3	SPIKE	70000	6000	76000
4	TYKE	62000	5500	67500

14) Show total and average budget per department.
Only include departments with average budget above 70000

```
SELECT department
        sum (budget) as total_budget,
        avg (budget) as avg_budget
FROM Projects
GROUP BY DEPARTMENT
HAVING AVG (budget) > 70000;
```


Results

Department	total budget	Avg budget
IT	27 0000	135 000
FINANCE	80 000	80 000

- 15) List all projects with budgets between 50 000 and 120 000, excluding the marketing department

```

SELECT Project_ID, Project_name, department,
        budget
FROM Projects
WHERE budget between 50 000 AND 120 000
AND DEPARTMENT <> 'Marketing';
  
```

RESULTS

Project ID	Project name	Department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	HR Portal	HR	50 000